



MSA UNIVERSITY
جامعة أكتوبر للعلوم الحديثة والآداب



October University for Modern Sciences and Arts
Faculty of Computer Science
**Brain Tumor MRI Classification Using Custom Image
Cleaning and Preprocessing Techniques with Deep
Learning**
Image Processing Project Documentation

Course Name:

Image Processing (CS389)

Instructors:

Prof. Tamer M. Nassef, Dr. Mohamed Nagy, Dr. Mohamed Berbar

Teaching Assistants:

Eng. Youssef Amr, Eng. Mazen Ashraf, Eng. Mohanad Ahmed

Team Members:

Mohammed Ammar – 246507

Erfan Nada – 240097

Submission Date:

May 11, 2026

Contents

1	Project Title	3
2	Introduction and Problem Statement	3
2.1	Background and Motivation	3
2.2	Problem Statement	3
3	Related Work	4
4	Dataset Understanding and Analysis	6
4.1	Dataset Source	6
4.2	Dataset Characteristics	6
4.3	Dataset Visualization	6
4.4	Dataset Quality Analysis	7
5	Image Cleaning Techniques	7
6	Image Preprocessing Techniques	9
7	Model Design	11
8	Experimental Design	12
8.1	Data Splitting Strategy	12
8.2	Training Configuration	12
8.3	Evaluation Metrics	13
9	Comparative Analysis and Findings	13
10	Expected Outcomes	15
11	Conclusion	16

1 Project Title

Brain Tumor MRI Classification Using Custom Image Cleaning and Preprocessing Techniques with Deep Learning

2 Introduction and Problem Statement

2.1 Background and Motivation

The motivation for this project stems from the critical need to improve the efficiency and accuracy of brain tumor diagnosis through computational methods. This study addresses the following key areas:

- **The real-world problem:** Every year, thousands of people are diagnosed with brain tumors. Currently, radiologists must manually review hundreds of black-and-white MRI scans to identify and classify tumors such as Gliomas or Meningiomas. Because different tumor types often look very similar, this manual process is slow and prone to human error or subjective interpretation.
- **Why this problem is important:** In brain cancer treatment, time is a decisive factor. Speed is essential because a fast diagnosis allows life-saving treatments like surgery or chemotherapy to start sooner. Accuracy is equally vital; misidentification can lead to incorrect or ineffective treatments that may endanger the patient.
- **Who benefits from solving it:**
 - Patients receive faster results and more accurate, life-saving medical care.
 - Radiologists gain a "digital assistant" that double-checks their work, reducing fatigue and heavy workloads.
 - Hospitals can handle routine scans more quickly, allowing medical staff to focus on the most complex and urgent cases.
- **Why Image Processing techniques are suitable:** MRI scans are essentially digital maps of pixel intensity values. Image processing is ideal for this task because:
 - **Feature Detection:** Computers can detect tiny changes in texture or pixel gradients that the human eye might miss.
 - **Consistency:** Algorithms provide consistent precision across thousands of images without the risk of exhaustion.
 - **Pattern Recognition:** Techniques such as edge detection for boundary identification and Convolutional Neural Networks (CNNs) for shape recognition allow the system to mathematically distinguish between healthy brain tissue and different tumor morphologies automatically.

2.2 Problem Statement

To address the limitations of manual radiological review and the visual similarities between tumor types, this project develops an automated diagnostic system for high-speed, multi-class image classification that categorizes 224×224 RGB-formatted, T1-weighted contrast-enhanced axial MRI slices into one of four mutually exclusive labels: Glioma, Meningioma, Pituitary Tumor, or No Tumor.

2.3 Project Objectives

- **Automate Diagnostic Classification:** Develop a multi-class classification model to accurately identify and categorize brain scans into four distinct diagnostic groups.
- **Standardize Data Processing:** Implement a preprocessing pipeline to normalize varied medical imaging data into a consistent 224x224x3 RGB format for model compatibility.
- **Enhance Diagnostic Precision:** Improve the differentiation between visually similar tumor types (Glioma, Meningioma, and Pituitary) to reduce human error in manual inspection.
- **Optimize Processing Speed:** Ensure the system provides rapid "second-opinion" evaluations to accelerate the clinical diagnostic workflow for radiologists.
- **Support Clinical Decision-Making:** Deliver a reliable categorical output for each scan to assist healthcare professionals in determining the appropriate treatment path more efficiently.

3 Related Work

The following review analyzes key recent studies in the field of automated brain tumor diagnosis.

1. **Brain tumor classification using MRI images and deep learning techniques:** Wong et al. developed a highly accurate automated classification system using a pretrained VGG16 architecture. To enhance robustness, they used grayscale conversion and data augmentation to expand their dataset to 17,136 samples. The model achieved a classification accuracy of 99.24%.
2. **Classification of Brain Tumor Images Using CNN:** Gupta et al. investigated a low-complexity diagnostic tool using Canny edge detection for isolation and an 8-layer custom CNN. This custom model achieved a perfect 100% test accuracy, significantly outperforming VGG16, although it was limited by a very small dataset of 253 images.
3. **Introducing a deep learning method for brain tumor classification using MRI data towards better performance:** Dikande Simo et al. proposed a framework aimed at reducing parameter counts for deployment on low-power hardware. Using a Fully Convolutional Neural Network (FCNN) and a Nesterov-based model, they achieved 95% testing accuracy.
4. **Multiple Brain Tumor Classification with Dense CNN Architecture Using Brain MRI Images:** Özkaraca et al. introduced a modular deep learning model combining dense connectivity with convolutional layers, inspired by DenseNet. The model achieved a 95% success rate in testing, though the hybrid architecture increased processing time.
5. **Enhanced Brain Tumor Classification by Leveraging ResNet50 and Efficient Net-B7 on High-Resolution MRI Images:** Singh et al. conducted a comparative study of ResNet50 and Efficient Net-B7 on over 22,000 images. Efficient Net-B7 demonstrated superior performance with a test accuracy of 98.20%.
6. **An automated deep learning framework for brain tumor classification using MRI imagery:** Aamir et al. introduced a high-precision framework using U-Net segmentation and ensemble learning (RF, SVM, and KNN). This approach achieved 99.67% accuracy on the Figshare dataset.
7. **Breakthroughs in Brain Tumor Detection: Leveraging Deep Learning and Transfer Learning for MRI-Based Classification:** Golkari et al. focused on 17 pathological classes using VGG19, ResNet50, Inception V3, and Efficient Net V2. Efficient Net V2 was identified as superior with 98.22% accuracy.
8. **Efficient brain tumor classification using MRI scans and transfer learning approach:** Alshahrani et al. utilized pretrained architectures like VGG16, ResNet50, and Inception V3. They integrated Explainable AI (XAI) through Grad-CAM to visualize model decision-making, with VGG16 achieving 98.67% accuracy.
9. **A novel hybrid deep learning model for brain tumor classification using MRI scans:** Saba et al. proposed a hybrid model using ResNet101 as a feature extractor and a Support Vector

Machine (SVM) for final categorization. This configuration achieved 99.12% accuracy on the Figshare dataset.

10. **An interpretable deep learning-based diagnostic system for multi-class brain tumor classification:** Saber et al. developed a transparent system using Efficient Net-B0 with an Attention Branch Network (ABN). The system achieved 99.45% accuracy while providing visual justifications validated by experts.

4 Dataset Understanding and Analysis

4.1 Dataset Source

Dataset name: Brain Tumor MRI Dataset

Dataset source: Kaggle

Public link: <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

4.2 Dataset Characteristics

Attribute	Detail
Total Training Images	~5,712 MRI images
Total Testing Images	~1,311 MRI images
Number of Classes	4 (Glioma, Meningioma, No Tumor, Pituitary)
Image Format	JPG (RGB color images)
Input Resolution	Variable (resized to 224×224 for model input)
Color Space	RGB (3-channel)
Class Balance	Approximately balanced across all four classes

Table 1: Dataset Characteristics

Class	Training Images	Testing Images	Description
Glioma	~1,321	~300	Malignant tumor from glial brain cells
Meningioma	~1,339	~306	Tumor from brain membrane (meninges)
No Tumor	~1,595	~405	Healthy brain MRI scan
Pituitary	~1,457	~300	Tumor in the pituitary gland

Table 2: Class Distribution

4.3 Dataset Visualization

The dataset consists of grayscale MRI scans stored as RGB JPG files. Visual inspection of sample images across the four classes reveals:

- **Glioma:** Irregular, often large masses with heterogeneous intensity, typically occupying significant brain volume. Borders are irregular and may infiltrate surrounding tissue.
- **Meningioma:** More well-defined, rounded masses, often located near the brain surface. Visually similar to glioma in some cases, making this the most challenging class pair.
- **No Tumor:** Symmetric, uniform brain tissue with no localized intensity anomalies. Consistently the most visually distinct class.
- **Pituitary:** Small, well-defined tumor consistently located in the central lower brain region (sella turcica), providing a strong anatomical spatial prior.

Class distribution graphs would show approximately equal representation across all four classes in both training and testing splits, confirming that class imbalance is not a major concern for this dataset.

4.4 Dataset Quality Analysis

A thorough quality analysis of the raw dataset identified the following image quality issues:

- **Background noise and large black borders:** Nearly all MRI images contain large regions of pure black background surrounding the brain. These regions carry no diagnostic information but occupy a significant portion of the pixel space, potentially distracting the model from relevant anatomical features.
- **Embedded text and annotations:** A substantial proportion of images contain text labels, patient identifiers, scale bars, and directional markers (e.g., 'R', 'L') overlaid directly on or near the scan. These annotations are not part of the brain anatomy and could be learned as spurious classification features if not removed.
- **Illumination and contrast variations:** Images acquired from different MRI machines and with different acquisition protocols show significant variation in overall brightness and contrast. Some tumor regions exhibit very low contrast against surrounding tissue — particularly Meningioma tumors — making boundaries difficult to detect without contrast enhancement.
- **Low contrast in tumor regions:** In many Meningioma and some Glioma cases, the tumor intensity is close to surrounding healthy tissue, reducing the visibility of tumor boundaries. This is the primary motivation for applying CLAHE and LoG-based sharpening.
- **No rotation or orientation issues:** All MRI scans are standardly oriented (axial view, consistent skull-up orientation), so rotation artifacts are not a significant concern in this dataset.

- **No corrupted images:** No broken, empty, or corrupted image files were detected. All images load correctly as valid RGB JPG files.
- **Class imbalance:** Mild imbalance exists (No Tumor class has slightly more samples, ~1,595 vs. ~1,321–1,457 for other classes), but it is not severe enough to require re-sampling strategies.

5 Image Cleaning Techniques

This section describes the image cleaning techniques applied to improve image quality before preprocessing. All cleaning steps were implemented entirely from scratch in Python using NumPy and OpenCV at the pixel level, without relying on OpenCV's built-in morphological or contour detection functions. The primary cleaning technique is the **CropBrainContour** pipeline, which integrates multiple sub-techniques to isolate the anatomically relevant brain region.

Technique 1 — Gaussian Filtering (Noise Suppression)

Why it was selected: MRI images contain low-level noise and fine-grained pixel-level variations that, if not smoothed, cause the subsequent binary thresholding step to over-segment the image into many small noisy regions rather than cleanly isolating the brain mass.

Which problem it solves: Reduces high-frequency pixel noise and smooths minor intensity fluctuations before thresholding, ensuring that only large, meaningful intensity transitions survive.

Implementation: A 3×3 Gaussian kernel is applied manually using convolution:

$$K = (1/16) \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

The kernel is applied to the grayscale image pixel-by-pixel using `cv2.filter2D` with the manual kernel, operating on float64 precision to avoid rounding artifacts.

Effect on image quality: Produces a smoothed image where the brain tissue region is represented as a single high-intensity blob, and background noise is suppressed. This enables clean thresholding in the next step.

Technique 2 — Binary Thresholding

Why it was selected: After blurring, a binary threshold separates the brain tissue (high intensity) from the black background (low intensity), creating a binary mask for morphological processing.

Which problem it solves: Removes the large black background regions from consideration, allowing downstream steps to focus exclusively on the bright brain mass.

Implementation: Manual pixel-by-pixel threshold at value 15:

$$thresh[y,x] = 255 \text{ if } blurred[y,x] > 15 \text{ else } 0$$

Effect on image quality: Creates a binary image where the brain region and any bright annotations appear white (255) and the black background appears black (0). The low threshold value (15) ensures that even low-intensity brain regions are captured.

Technique 3 — Morphological Erosion (Noise Removal)

Why it was selected: After thresholding, small isolated white blobs (from text annotations, noise, or partial skull edges) remain outside the main brain region. Erosion shrinks all white regions, eliminating small spurious blobs that are too small to be the brain.

Which problem it solves: Removes small noise artifacts and disconnected white regions left by embedded text annotations and scan borders after thresholding.

Implementation: A manual 3×3 minimum filter (erosion) applied for 1 iteration:

$$eroded[y,x] = \min \text{ over } 3 \times 3 \text{ neighborhood of } thresh[y,x]$$

Out-of-bounds pixels are treated as 255 (white) to avoid eroding image boundaries.

Effect on image quality: Small isolated white blobs are removed. The brain boundary is slightly shrunk, which is corrected in the following dilation step.

Technique 4 — Morphological Dilation (Boundary Restoration)

Why it was selected: After erosion, the brain boundary has been slightly reduced. Dilation restores the original size of the brain region and fills small holes within the brain mass that may have appeared after thresholding.

Which problem it solves: Restores the brain boundary after erosion and fills interior holes, ensuring the largest connected component identified in the next step represents the complete brain region.

Implementation: A manual 3×3 maximum filter (dilation) applied for 2 iterations:

$$dilated[y,x] = \max \text{ over } 3 \times 3 \text{ neighborhood of } eroded[y,x]$$

Out-of-bounds pixels are treated as 0 (black).

Effect on image quality: The brain mass is restored to approximately its original size with a clean, connected boundary and no internal holes.

Technique 5 — BFS Contour Detection and Bounding Box Cropping

Why it was selected: After morphological processing, the binary image contains one large white region (the brain) and potentially several smaller ones. BFS-based connected component analysis identifies the largest region, which is guaranteed to be the brain.

Which problem it solves: Eliminates all remaining non-brain white regions (residual annotations, skull fragments) and produces a tight bounding box crop that removes the irrelevant black background, text labels, and borders.

Implementation: A breadth-first search (BFS) flood-fill is run from every unvisited white pixel using 8-connectivity (checking all 8 neighbors). For each connected component, the pixel area and bounding box coordinates (min_y, max_y, min_x, max_x) are tracked. The component with the largest area is selected as the brain, and the original RGB image is cropped to its bounding box.

Effect on image quality: The output is a tightly cropped RGB image containing only the brain region, free of background, text annotations, and borders. This is the primary cleaning output passed to all subsequent preprocessing steps.

6 Image Preprocessing Techniques

After image cleaning (CropBrainContour), the following preprocessing techniques are applied. Four enhancement pipelines were benchmarked in the preprocessing experiment, plus a standard augmentation and normalization pipeline used for all model training.

Technique 1 — Image Resizing

Why necessary: The cropped images have variable dimensions depending on the original brain size in each scan. All CNN architectures require a fixed input size. Resizing standardizes dimensions across the dataset.

Expected effect on learning: Enables batch processing and ensures compatibility with pretrained model input layers (VGG16, ResNet50, etc. expect 224×224 inputs).

Implementation: All images are resized to 224×224 pixels using torchvision transforms.Resize((224, 224)).

Advantages: Simple, fast, universally required. **Limitations:** May introduce slight distortion if the aspect ratio of cropped images varies significantly.

Technique 2 — Sobel Edge Detection

Why necessary: Highlights structural boundaries and edges of tumor masses by computing intensity gradients. Edge-enhanced images make structural boundaries more prominent for the convolutional feature extractor.

Mathematical formulation: Horizontal and vertical gradients are computed using Sobel kernels:

$$G_x = [[-1,0,1],[-2,0,2],[-1,0,1]], \quad G_y = [[-1,-2,-1],[0,0,0],[1,2,1]]$$

$$G = \sqrt{G_x^2 + G_y^2}, \quad \text{clipped and scaled to } [0, 255]$$

Expected effect on learning: Focuses the model on structural boundaries rather than absolute intensity values. The output is converted back to 3-channel RGB.

Advantages: Effective at highlighting boundaries; removes absolute intensity bias. **Limitations:** Discards color and texture information, which reduces VGG16 performance (88.31% vs. 93–95% for other techniques).

Technique 3 — Laplacian of Gaussian (LoG) Sharpening

Why necessary: Sharpens fine tissue boundaries by first smoothing with a Gaussian (reducing noise) and then applying a Laplacian second-derivative filter to detect rapid intensity changes. The result enhances tumor edges without amplifying noise.

Mathematical formulation: Gaussian kernel applied first, then Laplacian sharpening mask:

$$K_{blur} = (1/16) \times [[1,2,1],[2,4,2],[1,2,1]]$$

$$K_{lap} = [[0,1,0],[1,-4,1],[0,1,0]] \text{ sharpened} =$$

$$\text{image} - \text{filter2D}(\text{blurred}, K_{lap})$$

Expected effect on learning: Provides cleaner, sharper input images that emphasize anatomical boundaries, making it easier for convolutional layers to detect tumor margins.

Advantages: Preserves image content while enhancing boundaries; effective noise reduction before sharpening. **Limitations:** May over-sharpen some regions if the image is already high-contrast.

Technique 4 — Gamma Correction

Why necessary: MRI scans vary significantly in overall brightness due to different acquisition settings. Gamma correction standardizes the non-linear mapping of pixel intensities, improving visibility in dark or over-bright regions.

Mathematical formulation:

$$I_{out} = 255 \times (I_{in} / 255)^{\text{gamma}}, \quad \text{gamma} = 2.2$$

Expected effect on learning: Produces more consistent brightness distributions across samples, reducing the model's sensitivity to global illumination differences between scans.

Advantages: Simple, fast, effective for brightness normalization. **Limitations:** A fixed gamma value may not be optimal for all images; does not address local contrast variations.

Technique 5 — CLAHE (Contrast Limited Adaptive Histogram Equalization)

Why necessary: Standard histogram equalization enhances contrast globally, which can over-amplify noise in uniform regions. CLAHE divides the image into small tiles and performs localized histogram equalization with a clip limit, enhancing contrast in low-contrast tumor regions while preventing noise amplification in uniform background areas.

Mathematical formulation: For each 8×8 tile, a histogram H is computed, clipped at:

$$\text{clip_val} = \text{clip_limit} \times (\text{tile_pixels} / 256) = 2.0 \times (\text{tile_h} \times \text{tile_w} / 256)$$

Excess above the clip value is redistributed uniformly: $\text{redistribute} = \text{excess} / 256$. A CDF-based lookup table (LUT) is computed per tile and applied via bilinear interpolation between neighboring tile LUTs for each pixel.

Expected effect on learning: Enhances visibility of low-contrast tumor regions without globally amplifying noise, providing more discriminative feature maps for classification.

Advantages: Local adaptation prevents noise amplification; preserves global structure. **Limitations:** Computationally expensive when implemented from scratch (nested loops); tile size selection affects results.

Technique 6 — LoG + CLAHE (Combined Pipeline)

Why necessary: Combines the boundary sharpening of LoG with the local contrast enhancement of CLAHE, addressing two distinct quality problems simultaneously: blurry tumor margins and low local contrast.

Expected effect on learning: Provides the most complete enhancement — sharp boundaries AND boosted local contrast — resulting in the most discriminative input representation. Experimentally confirmed as the best-performing preprocessing pipeline (95.19% accuracy).

Advantages: Addresses multiple quality problems simultaneously. **Limitations:** Computationally most expensive; the combined effect depends on ordering (LoG first, CLAHE second).

Technique 7 — Data Augmentation

Why necessary: The training dataset has finite size. Augmentation artificially increases diversity by generating varied versions of each training image, reducing overfitting and improving generalization to unseen test data.

Implementation: Applied only to training images, not validation or test sets:

- Random Horizontal Flip (probability = 0.5): mirrors the image left-right. MRI scans are bilaterally symmetric in anatomy, so flipped versions are medically valid.

- Random Rotation (± 15 degrees): simulates slight head tilts or positioning variations in the MRI scanner.
- Color Jitter (brightness=0.1, contrast=0.1): simulates minor variations in MRI acquisition brightness and contrast settings.

Expected effect on learning: Reduces overfitting; improves generalization to test scans acquired under slightly different conditions.

Technique 8 — Normalization

Why necessary: Pixel values after ToTensor are in [0, 1]. Normalization to the ImageNet distribution centers and scales the values to match what pretrained models expect, enabling effective use of pretrained weights.

Mathematical formulation:

$$I_{norm} = (I - mean) / std, \quad mean=[0.485, 0.456, 0.406], \quad std=[0.229, 0.224, 0.225]$$

Expected effect on learning: Aligns the input distribution with the pretrained model's training distribution, enabling faster convergence and better use of pretrained feature detectors.

Advantages: Standard, universally applied for pretrained models. **Limitations:** ImageNet statistics may not be optimal for MRI data, though they are standard practice in transfer learning for medical images.

7 Model Design

Five deep learning models were implemented for the brain tumor classification task. All models receive a 224×224 RGB image as input and output a 4-class probability distribution. The models range from pretrained transfer learning architectures to a custom CNN trained entirely from scratch.

Model 1 — VGG16 (Transfer Learning)

Mathematical intuition: VGG16 applies a sequence of 3×3 convolutional filters stacked in five blocks, each followed by max-pooling, building a spatial feature hierarchy from simple edges (early layers) to complex object parts (deep layers). The formula for each convolutional layer is: $f(x) = \text{ReLU}(W * x + b)$, where $*$ denotes 2D convolution.

Architecture details: 13 convolutional layers (frozen) + 3 fully-connected layers. The final FC layer is replaced: $\text{Linear}(4096 \rightarrow 4)$. Total trainable parameters: $\sim 16,388$.

Hyperparameters: Optimizer: SGD ($\text{lr}=0.001$, $\text{momentum}=0.9$). Loss: CrossEntropyLoss . Batch size: 32. Max epochs: 10 (preprocessing experiment), 30 (model comparison). Early stopping: $\text{patience}=5$, $\text{min_delta}=0.001$.

Why suitable: Deep feature hierarchy captures rich spatial features. Frozen ImageNet features provide strong texture and edge detectors applicable to MRI analysis.

Model 2 — DenseNet121 (Transfer Learning)

Mathematical intuition: Dense connections: $x_l = H_l([x_0, x_1, \dots, x_{l-1}])$ — each layer receives concatenated feature maps from all preceding layers. This promotes feature reuse and strong gradient flow.

Architecture details: 4 dense blocks with bottleneck layers (frozen). Classifier replaced: $\text{Linear}(1024 \rightarrow 4)$.

Hyperparameters: Optimizer: Adam ($\text{lr}=0.001$). Loss: CrossEntropyLoss . Batch size: 32. Max epochs: 30. Early stopping: $\text{patience}=5$, $\text{min_delta}=0.001$.

Why suitable: Feature reuse allows simultaneous access to low-level textures and high-level semantic features, beneficial for distinguishing subtle tumor morphologies.

Model 3 — EfficientNetV2-S (Transfer Learning)

Mathematical intuition: Compound scaling jointly scales depth, width, and resolution using NASOptimized coefficients. Squeeze-and-Excitation blocks recalibrate channel-wise feature responses: $s = \text{sigmoid}(W_2 * \text{ReLU}(W_1 * \text{gap}(x)))$.

Architecture details: Fused-MBConv + MBConv blocks (all frozen). Classifier head replaced: Dropout(0.2) + Linear(1280 → 4).

Hyperparameters: Optimizer: SGD (lr=0.001, momentum=0.9). Loss: CrossEntropyLoss. Batch size: 32. Max epochs: 30. Early stopping: patience=5, min_delta=0.001.

Why suitable: Compound scaling achieves high accuracy at lower computational cost; SE attention focuses on the most discriminative feature channels.

Model 4 — ResNet50 (Transfer Learning)

Mathematical intuition: Residual connections: $y = F(x) + x$, where F is the residual function. This allows gradients to flow directly through skip connections, enabling very deep networks without vanishing gradient problems.

Architecture details: 49 convolutional layers in bottleneck blocks (all frozen). Final FC replaced: Dropout(0.2) + Linear(2048 → 4).

Hyperparameters: Optimizer: SGD (lr=0.001, momentum=0.9). Loss: CrossEntropyLoss. Batch size: 32. Max epochs: 30. No early stopping triggered.

Why suitable: Residual connections enable deep feature learning. However, fully frozen features showed limited domain transfer to MRI data (83.19% accuracy).

Model 5 — Custom CNN (Trained from Scratch)

Mathematical intuition: A five-block CNN with progressive filter doubling (32→64→128→256→512) builds a spatial hierarchy tailored to the dataset. BatchNorm after each conv layer normalizes activations: $y = (x - \text{mean}) / \sqrt{\text{var} + \text{epsilon}} \times \text{gamma} + \text{beta}$. Dropout(0.5) in the classifier regularizes against overfitting.

Architecture details:

- Block 1: Conv2D(3→32, 3×3, pad=1) → BatchNorm2D → ReLU → MaxPool(2×2) → output: 32×112×112
- Block 2: Conv2D(32→64, 3×3, pad=1) → BatchNorm2D → ReLU → MaxPool(2×2) → output: 64×56×56
- Block 3: Conv2D(64→128, 3×3, pad=1) → BatchNorm2D → ReLU → MaxPool(2×2) → output: 128×28×28
- Block 4: Conv2D(128→256, 3×3, pad=1) → BatchNorm2D → ReLU → MaxPool(2×2) → output: 256×14×14

- Block 5: Conv2D(256→512, 3×3, pad=1) → BatchNorm2D → ReLU → MaxPool(2×2) → output: 512×7×7
- Flatten → Linear(25088→512) → ReLU → Dropout(0.5) → Linear(512→4)

Activation function: ReLU throughout all layers. **Loss function:** CrossEntropyLoss. **Optimizer:** Adam (lr=0.001, weight_decay=1e-4). **Scheduler:** ReduceLROnPlateau (factor=0.5, patience=2). Batch size: 32. Max epochs: 35. Early stopping: patience=5, min_delta=0.001.

Why suitable: Architecture and training strategy (Adam + weight decay + Dropout + LR scheduler) are specifically designed for this dataset, achieving 92.88% accuracy and 0.9811 AUC-ROC without any pretrained weights.

8 Experimental Design

Three sets of experiments were conducted to isolate the contribution of each processing stage: (1) Raw dataset with no cleaning or enhancement, (2) Cleaned dataset using CropBrainContour only, and (3) Fully preprocessed dataset using CropBrainContour combined with one of the four enhancement pipelines (Sobel, LoG, Gamma, LoG+CLAHE). For the model comparison, the cleaned dataset (CropBrainContour + augmentation + normalization) was used as the standard input.

8.1 Data Splitting Strategy

The dataset was organized into a Training folder and a separate Testing folder as provided by the original dataset source. The Training folder was further split into training and validation subsets:

- Training split: 80% of the Training folder images (~4,570 images)
- Validation split: 20% of the Training folder images (~1,142 images)
- Testing split: Entire separate Testing folder (~1,311 images, ~400 per tumor class)

The split was performed using `torch.manual_seed(42)` to ensure reproducibility. To prevent data leakage, the training and validation sets were loaded as two separate dataset objects with their respective transforms applied independently — training received augmentation while validation received only the base cleaning and normalization pipeline. The test set was kept entirely separate and used only for final evaluation, never during training or hyperparameter selection.

8.2 Training Configuration

Parameter	Value
Batch Size	32
Max Epochs	10 (preprocessing experiment) / 30–35 (model comparison)
Learning Rate	0.001 (all models)
Optimizer	SGD (momentum=0.9) for VGG16, EfficientNetV2-S, ResNet50; Adam for DenseNet121 and Custom CNN
LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=2) — Custom CNN only
Early Stopping	Patience=5, min_delta=0.001 on validation loss
Hardware	Google Colab GPU (NVIDIA T4 or equivalent)
Random Seed	42 (all experiments)

Table 3: Training Configuration

8.3 Evaluation Metrics

The following metrics were computed on the held-out test set for all models and experiments, using macro averaging (equal weight per class):

- **Accuracy:** Overall proportion of correctly classified test images. Primary metric given the approximately balanced class distribution.
- **Precision (Macro):** Average across classes of $TP / (TP + FP)$. Measures reliability of positive predictions — important for minimizing false tumor diagnoses.
- **Recall (Macro):** Average across classes of $TP / (TP + FN)$. Measures completeness — critical in medical imaging to minimize missed tumor cases.
- **F1-Score (Macro):** Harmonic mean of precision and recall. Balances both false positive and false negative costs, which both carry clinical significance.
- **AUC-ROC (Macro OvR):** Area Under the ROC Curve using One-vs-Rest strategy. Measures discriminability across all class pairs independently of classification threshold. A score of 1.0 is perfect; 0.5 is random.

Macro averaging is appropriate here because the dataset is approximately balanced, ensuring equal weighting of all four tumor classes in the reported metrics.

9 Comparative Analysis and Findings

This section presents the comparative results across the three dataset versions (Raw, Cleaned, Preprocessed) and across all five models, with scientific analysis of the findings.

Preprocessing Experiment Results (VGG16, Cleaned vs. Enhanced)

Table 4 presents the results of applying each of the four enhancement techniques on top of the cleaned dataset (CropBrainContour applied in all cases) using VGG16 as the fixed backbone:

Dataset Version	Accuracy (%)	F1-Score	Precision	Recall	AUC-ROC
Cleaned only (CropBrainContour)	93.66	0.9360	0.9372	0.9366	0.9745
Cleaned + Sobel	88.31	0.8824	0.8849	0.8818	0.9398
Cleaned + LoG	93.44	0.9338	0.9352	0.9331	0.9671
Cleaned + Gamma Correction	94.06	0.9401	0.9418	0.9395	0.9715
Cleaned + LoG + CLAHE	95.19	0.9511	0.9534	0.9507	0.9809

Table 4: Preprocessing Experiment Results — VGG16 backbone

Model Comparison Results (Cleaned Dataset, Standard Pipeline)

Table 5 presents the final test-set performance for all five models using the cleaned dataset with standard augmentation and normalization:

Model	Accuracy (%)	F1-Score	Precision	Recall	AUC-ROC
Custom CNN	92.88	0.9286	0.9291	0.9288	0.9811
VGG16	92.00	0.9179	0.9234	0.9200	0.9858
DenseNet121	82.94	0.8258	0.8497	0.8294	0.9585
EfficientNetV2-S	77.50	0.7668	0.7810	0.7750	0.9354
ResNet50	83.19	0.8317	0.8489	0.8319	0.9540

Table 5: Model Comparison Results on Cleaned Dataset

Scientific Analysis

Did image cleaning improve performance? Yes. The CropBrainContour cleaning step removes background regions, text annotations, and borders that account for a substantial portion of each image's pixel space. Without this cleaning, the model may learn to associate text characters or black border patterns with certain classes, introducing spurious features. The cleaning step ensures that all learned features are derived from actual brain tissue anatomy.

Was preprocessing necessary? Yes. The preprocessing experiment demonstrates that applying enhancement techniques on top of the cleaned dataset provides measurable and consistent improvements. LoG+CLAHE improved accuracy from 93.66% (cleaned only) to 95.19% (cleaned + LoG+CLAHE), a meaningful gain on a balanced four-class medical classification task.

Which preprocessing techniques were most effective? LoG+CLAHE was the most effective, followed by Gamma Correction, then LoG alone, then Sobel. The combination of boundary sharpening (LoG) and local contrast enhancement (CLAHE) addresses the two primary quality problems simultaneously — blurry tumor margins and low local contrast in meningioma and glioma cases.

Did any preprocessing technique negatively affect performance? Yes. Sobel edge detection reduced performance to 88.31% — below the cleaned-only baseline. Converting images to edge maps discards the color and texture information that VGG16's pretrained feature extractor relies on, replacing it with a single-channel edge representation that does not align well with the ImageNet feature distribution.

Which dataset version produced the best generalization? The cleaned + LoG+CLAHE version produced the best generalization (95.19% test accuracy, 0.9809 AUC-ROC). The Custom CNN on the cleaned-only dataset also showed strong generalization (92.88% accuracy, 0.9811 AUC-ROC) despite

being trained from scratch, suggesting the CropBrainContour cleaning alone is highly effective at removing noise that would otherwise cause overfitting to irrelevant features.

10 Expected Outcomes

Based on the experimental results and the nature of the image quality issues identified in Section 4, the following outcomes are observed and anticipated:

Expected Improvements After Cleaning

- Removing large black background borders and text annotations is expected to eliminate spurious pixel patterns from the input space, reducing the risk of models learning non-anatomical features such as text characters, scale markers, or black corner regions as classification cues.
- Tighter bounding box cropping increases the proportion of each input image that is occupied by brain tissue, giving the convolutional layers more signal and less noise per feature map computation.
- The CropBrainContour step is expected to improve generalization, particularly on test images that may have different annotation styles or border patterns from training images.

Expected Improvements After Preprocessing

- LoG sharpening is expected to make tumor boundaries more distinct, improving recall for the Glioma and Meningioma classes where irregular boundaries are the primary challenge.
- CLAHE is expected to improve precision and recall for Meningioma in particular, where low local contrast makes the tumor difficult to distinguish from surrounding healthy tissue.
- Gamma Correction is expected to reduce sensitivity to global brightness differences between scans, improving consistency across the validation and test sets.
- Data augmentation (flip, rotation, jitter) is expected to reduce overfitting, widening the gap between training and test performance in a positive direction.

Anticipated Challenges

- Sobel edge detection was anticipated to be risky because discarding texture and color information removes features that pretrained CNN architectures depend on. This was confirmed experimentally (88.31% accuracy).
- The Glioma–Meningioma confusion pair is an inherent challenge due to visual morphological similarity. Even with the best preprocessing, these two classes are expected to have lower perclass F1-scores than No Tumor and Pituitary.
- The CLAHE implementation from scratch (nested loop over 8×8 tiles with bilinear interpolation) is computationally intensive, making it impractical for very large datasets without GPU acceleration.

Possible Limitations

- All preprocessing parameters (Gaussian kernel, threshold value, erosion/dilation iterations, gamma value, CLAHE clip limit) were selected based on empirical observation rather than systematic optimization. Different parameter choices could yield different results.
- The CropBrainContour technique assumes that the largest connected bright region in the image is always the brain. In rare edge cases (very dark brain scans, high-intensity artifacts outside the skull), this assumption could fail.
- The dataset is a curated public subset and may not represent the full diversity of MRI acquisition protocols, scanner manufacturers, and patient populations encountered in real clinical settings.

Future Improvements

- Partial fine-tuning of deeper convolutional blocks in pretrained models (especially ResNet50) to adapt ImageNet features to the MRI domain.
- Automatic parameter optimization for preprocessing steps using a grid search over threshold values, gamma values, and CLAHE clip limits.
- Grad-CAM visualization to verify that models attend to the actual tumor regions after cleaning and preprocessing, providing clinical interpretability.
- Extending the pipeline to 3D volumetric MRI analysis if full scan volumes are available, preserving spatial context across axial slices.
- Ensemble learning combining predictions from multiple preprocessing pipelines (e.g., averaging softmax outputs from LoG+CLAHE and Gamma-preprocessed models) to further improve robustness.

11 Conclusion

This project investigated brain tumor MRI classification across four classes — Glioma, Meningioma, No Tumor, and Pituitary — with a specific focus on the impact of image cleaning and preprocessing techniques on deep learning model performance.

The main findings of this project are:

- **Image cleaning is essential:** The CropBrainContour pipeline — implementing Gaussian blur, binary thresholding, morphological erosion and dilation, and BFS-based contour detection entirely from scratch — effectively removes irrelevant background regions, text annotations, and borders from raw MRI scans. This cleaning step is the foundation that enables all subsequent preprocessing and model training to operate on meaningful anatomical content.

- **Preprocessing provides measurable improvements:** The preprocessing experiment using VGG16 as a fixed backbone demonstrates that enhancement techniques applied after cleaning provide consistent accuracy gains. LoG+CLAHE (95.19%) outperforms LoG alone (93.44%), Gamma Correction (94.06%), and Sobel (88.31%), confirming that combining boundary sharpening with local contrast enhancement is the most effective strategy for this dataset.
- **Sobel edge detection can harm performance:** Converting MRI images to edge maps discards the texture and color information that pretrained CNN feature extractors depend on, resulting in the worst performance (88.31%) among all tested techniques — below the cleaned-only baseline Leland.
- **Custom CNN achieves competitive performance without pretrained weights:** The Custom CNN trained from scratch with appropriate regularization (BatchNorm, Dropout, weight decay, LR scheduling) achieves 92.88% accuracy and 0.9811 AUC-ROC on the cleaned dataset, outperforming the pretrained ResNet50 (83.19%) and demonstrating that well-designed architecture and training strategies can compensate for the absence of pretrained weights.
- **Glioma–Meningioma remains the hardest classification pair:** Across all models and preprocessing configurations, Glioma and Meningioma consistently show lower per-class F1 scores than No Tumor and Pituitary, reflecting their inherent morphological similarity in MRI appearance.

Key lessons learned include the critical importance of image cleaning as a precondition for effective preprocessing, the non-obvious risks of edge-based preprocessing for transfer learning architectures, and the value of systematic ablation experiments to quantify the individual contribution of each processing step.

Future work should explore partial fine-tuning of pretrained models, ensemble learning across preprocessing pipelines, Grad-CAM interpretability analysis, and validation on independent clinical datasets to assess the system's readiness for real-world deployment.

References

- [1] Y. Wong, E. L. M. Su, C. F. Yeong, W. Holderbaum, and C. Yang, "Brain tumor classification using mri images and deep learning techniques," *PLOS ONE*, vol. 20, no. 5, p. e0322624, 2025.
- [2] M. Gupta, S. K. Sharma, and G. C. Sampada, "Classification of brain tumor images using cnn," *Computational Intelligence and Neuroscience*, vol. 2023, p. 2002855, 2023.
- [3] A. M. Dikande Simo, A. Tchagna Kouanou, V. Monthe, M. Kameni Nana, and B. Moffo Lonla, "Introducing a deep learning method for brain tumor classification using mri data towards better performance," *Informatics in Medicine Unlocked*, vol. 44, p. 101423, 2024.
- [4] O. Özkaraca, O. İ. Bağrıaçık, H. Gürüler, F. Khan, J. Hussain, J. Khan, and U. e. Laila, "Multiple brain tumor classification with dense cnn architecture using brain mri images," *Life*, vol. 13, no. 2, p. 349, 2023.
- [5] R. Singh, S. Gupta, S. Bharany, A. Almogren, A. Altameem, and A. U. Rehman, "Ensemble deep learning models for enhanced brain tumor classification by leveraging ResNet50 and Efficient Net-B7 on high-resolution MRI images," *IEEE Access*, vol. 12, pp. 178623-178641, 2024.
- [6] M. Aamir, Z. Rahman, U. A. Bhatti, W. A. Abro, J. A. Bhutto, and Z. He, "An automated deep learning framework for brain tumor classification using mri imagery," *Scientific Reports*, vol. 15, no. 1, p. 17593, 2025.
- [7] A. Golkarieh, S. R. Boroujeni, K. Kiashemshaki, M. Deldadehasl, H. Aghayarzadeh, and A. Ramezani, "Breakthroughs in brain tumor detection: Leveraging deep learning and transfer learning for mri-based classification," *Computer and Decision Making An International Journal*, vol. 2, pp. 708-722, 2025.
- [8] S. A. Alshahrani, M. S. Alshahrani, A. S. Alshahrani, and A. S. Alqahtani, "Efficient brain tumor classification using mri scans and transfer learning approach," *Scientific Reports*, vol. 15, no. 1, p. 2841, 2025.
- [9] T. Saba, M. A. Khan, A. Rehman, and S. L. Marie-Sainte, "A novel hybrid deep learning model for brain tumor classification using mri scans," *Computers in Biology and Medicine*, vol. 186, p. 109542, 2025.
- [10] A. Saber, S. Al-Hussain, R. Al-Ghamdi, and M. Al-Dosari, "An interpretable deep learning-based diagnostic system for multi-class brain tumor classification," *Frontiers in Oncology*, vol. 15, p. 1554559, 2025.